

A phase field approach to the fracture simulation of quasi-brittle structures with initial state

Caihong Zhang^c, Jiaming Zhang^a, Hanzhang Li^d, Yu Li^e, Wenqiang Xu^{a,b,*},
Sheng Qiang^{a,*}, Min Yuan^{f,a}

^a College of Water Conservancy and Hydropower, Hohai University, Nanjing, China

^b College of Water Resources and Civil Engineering, Hunan Agricultural University, Changsha, China

^c China Construction Seventh Engineering Division Corp. Ltd, Zhengzhou, China

^d College of Civil Engineering, Tongji University, Shanghai, China

^e Department of Architecture and Civil Engineering, City University of Hong Kong, Hong Kong, China

^f Chang Jiang Scientific Research Institute, Wuhan, China

ARTICLE INFO

Keywords:

Phase field method

Initial state

Crack propagation

Initial equilibrium iteration

Adaptive refinement mesh

ABSTRACT

To solve the simulation problem of crack propagation in solid structures with initial stress, this paper proposes a phase field approach that considers the initial state of the quasi-brittle structure. By using the initial equilibrium iteration method, the estimated initial state of the structure, including initial stress, is transformed into internal stress loads of the structure. Thus, the problem of non-convergence caused by inaccurate initial state estimation of the structure in engineering calculations was solved, and an improved subproblem staggered iteration method was adopted to simulate crack propagation of the structure under external loads. The universality and computational accuracy of this method for different phase field models were verified by using pre made notched thin plates for the expansion of mode I and II cracks in different initial states. Among them, the calculation error using the PF_CZM model is within 1%, and the calculation error of the AT2 model increases with the initial stress, but is less than 7%. At the same time, the crack growth problem of cracked structures under different uniform initial stresses is accurately simulated, which reflects the applicability of this method to actual engineering. In addition, the study also verified the universal applicability of the adaptive refinement strategy proposed based on the AT2 model for other phase field models.

1. Introduction

Fracture is one of the main damage modes of solid structures, and simulating and predicting structural fracture is an important research content. Under the finite element framework, numerical simulation methods for crack propagation problems can be mainly divided into two categories. One type is geometric description methods, where cracks are expanded based on the mesh generated by meshing, such as element deletion methods [1] and interface element methods [2]. While the crack can only propagate on the mesh boundary, which is mesh dependence [3] and numerical instability [4]. Another type is non-geometric description methods, such as Extended Finite Element method (XFEM) [5,6] and phase field method (PFM) [7,8]. In this type of method, cracks can detach from the

* Corresponding authors at: College of Water Conservancy and Hydropower, Hohai University, Nanjing, China.

E-mail addresses: xwqiang@hhu.edu.cn (W. Xu), sqiang2118@hhu.edu.cn (S. Qiang).

<https://doi.org/10.1016/j.engfracmech.2024.110150>

Received 20 October 2023; Received in revised form 26 April 2024; Accepted 11 May 2024

Available online 13 May 2024

0013-7944/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

grid and expand inside the grid. XFEM is a discontinuous interface processing method and requires special crack tracking technology (such as Level Sets [9] and Fast Marching [10]) to track the crack surface. However, these methods are particularly cumbersome in dealing with three-dimensional and multi-crack problems. PFM diffuses the crack interface and does not need to explicitly track the crack surface during the calculation process. It is automatically obtained through the evolution of order parameters, so it can accurately handle multi-crack and three-dimensional crack expansion problems. Therefore, PFM is an important method for simulating crack propagation, and has received widespread attention and development in recent years.

The PFM based on the fracture energy variational principle can solve the problem of the Griffith energy method requiring a predetermined crack propagation path [8]. In theory, the elliptical regularization method based on the Mumford Shah functional [11] (AT2 model [12]) regularizes sharp cracks into crack zones with a certain scale, and introduces crack phase field variables (d) and their gradients that are continuously distributed within the range $[0, 1]$ to describe the crack state. Miehe et al. [7,13] introduced concepts such as energy tension–compression decomposition [14] and historical maximum damage energy release rate [15] in classical damage mechanics, and re-derived a brittle phase field model, which was successfully applied to static, dynamic, and multi field coupled failure simulations of solid structures. Marigo et al. [16] proposed the brittle phase field model AT1, which solved the problem of the lack of linear elastic segment in the AT2 model. Wu Jianying et al. [17,18] combined fracture mechanics and damage mechanics and introduced the crack geometry function and energy degradation function to respectively characterize the crack phase field distribution characteristics and the degradation law of the strain energy of the cracked solid during solid failure. Thus, a unified phase field theory suitable for brittle fracture and quasi-brittle failure was established [19]. The proposed phase field regularized cohesive zone model (PF-CZM) [20,21] has been successfully applied to the damage and failure analysis of various solid materials and structures such as concrete [19], rock [22], glass [23], PMMA [21] under static, dynamic, and multi field coupling conditions.

Although PFM has been widely used and developed, the situation in actual engineering is more complex and it cannot solve all problems. One of the common problems is that in actual projects, the structure already contains large initial stress before external loads are applied. For example, due to geological movements, underground rock structures often contain large crustal stresses [24,25]. The stress often exceeds the compressive stress caused by self-weight and cannot be ignored when simulating crack propagation. Steel melts during welding, forming large thermal stress during the cooling process [26,27]. If this part of the stress is ignored when calculating the structural bearing capacity, the structural bearing capacity will be overestimated, causing safety hazards. For early-age mass concrete, a large temperature gradient is formed due to hydration heat release, resulting in large thermal stress [28,29]. When simulating crack propagation during the operation of the structure, this part of the tensile stress cannot be discarded.

In previous studies, uniform Dirichlet boundary conditions are primarily imposed on the displacement field to apply uniform initial stresses to the structure [30,31]. However, applying initial stresses at the boundaries of the computational domain can lead to significant deformations at the stress boundaries, which are inconsistent with engineering observations, and thus does not adequately consider the impact of initial stresses on crack propagation. To address this, Shuwei Zhou et al. [32] have established a new energy functional to consider the effects of the initial in-situ stress field. Subsequently, this energy functional is used to implement the control equations for the displacement field and phase field through a variational method. However, the numerical implementation of this theory is challenging and may not be suitable for engineering applications. Enrico Salvati [33] employed the PFM coupled with eigenstrain theory to simulate crack propagation behavior and introduce a residual stress field in a pre-notched sample. The eigenstrain approach is self-consistent, satisfying both stress balance and strain compatibility, but the method is relatively complex, involves the computation of R-curves, and is not conducive to engineering applications. Moreover, the suitability of this method for complex problems remains to be studied. Therefore, there is a need for a PFM that can consider the initial state of quasi-brittle structures and is easy to implement, making the simulation results more consistent with actual conditions.

At the same time, due to the limitations of actual survey conditions, we cannot accurately predict the initial state of the quasi-brittle structure. For example, when conducting sampling surveys on underground structures, only the initial stress data of the sampling points can be obtained, and accurate initial stress data for all areas cannot be provided in detail. Especially for the presence of local fracture zones such as joints, cracks, and interlayers, it is difficult to accurately estimate the initial stress in these areas due to the phenomenon of stress concentration. If such estimated stress is used to directly simulate crack propagation, the calculation will often fail to converge because the strain compatibility conditions are not met. Therefore, when establishing PFM that considers the initial state of the structure, this possible situation needs to be dealt with to avoid the failure of the calculation results to converge after external loads are applied.

Of course, it cannot be ignored that in order to simulate different material properties, different phase field models need to be used. Therefore, the PFM needs to meet the applicability of different models, and the calculation error needs to be controlled within a certain range to meet the accuracy requirements required for engineering calculations. Furthermore, special cases in engineering where the initial strain or initial displacement of a structure may be provided without the initial stress should also be addressed, ensuring that the model has a certain level of universality. For practicality considerations, the model should not be overly complex to avoid impacting engineering applications.

Therefore, the main purpose of this paper is to explore how the existing PFM can be applied to take into account the initial state of quasi-brittle structures, which includes initial stresses, strains, and displacements. Through this application, an enhancement in the accuracy of simulating structural fractures is aimed for. The organizational framework is structured as follows: In Section 2, based on the unified phase-field theory and previous research [34], the application of the PFM is discussed in terms of considering the effects of the structural initial state, encompassing the basic theory, numerical implementation methods, and computational steps. Additionally, methods to address the issue of non-convergence caused by inaccurate initial state estimation are explored. In Section 3, by utilizing the classic case of a pre-notched thin plate, the universality of this method under different phase-field models and various initial stress conditions is demonstrated, and an analysis of computational errors to assess accuracy is provided. Furthermore, the applicability of

this method in engineering applications is validated through examples with uniformly estimated initial stresses. In Section 4, the suitability of this method for specific initial strain and displacement conditions is discussed, providing useful references for the PFM in simulating the crack propagation in actual engineering structures. The code package containing PFM framework is programmed and implemented on MATLAB. It can be obtained from the corresponding authors.

2. Methodology

2.1. Basic theory of PFM

After years of development, the basic theory of the phase field method (PFM) has become quite mature. Based on the unified phase field theory proposed by Wu Jianying et al. [17,18] and combined with previous research [34,35], this article provides a phase field model that considers the initial state of the solid structure.

According to the image segmentation theory [36], the crack area A_s can be expressed as the following volume fraction [13]:

$$A_s = \underbrace{\int_S dA}_{\text{Sharp crack}} = \underbrace{\int_B \gamma(d; \nabla d)}_{\text{Regularized crack}} = A_d(d) \quad (1)$$

where S is the sharp crack surface, B is the crack zone, which means converting the sharp crack S into a finite-scale crack zone B , and introducing the crack phase field d to describe its state. $d \in [0, 1]$, where $d = 0$ means that the structure is intact, and $d = 1$ means that the structure is completely destroyed. ∇d is the phase field gradient. b is the crack density functional. Ambrosio et al. [37] gave the following form:

$$\begin{cases} \gamma(d; \nabla d) = \frac{1}{c_a} \left[\frac{1}{l} \alpha(d) + l |\nabla d|^2 \right] \\ c_a = 4 \int_0^1 \sqrt{\alpha(\beta)} d\beta \end{cases} \quad (2)$$

where $\alpha(d)$ is the crack geometry function, which is an increasing function of the phase field d , and its value range is $[0, 1]$. Its functional form can be:

$$\alpha(d) = \xi d + (1 - \xi) d^2 \quad (3)$$

where ξ is an adjustable parameter, and its value range is $[0, 2]$.

According to the first and second laws of thermodynamics, and based on the irreversibility of the evolution of the crack phase field, the governing equation of the phase field in the computational domain (Ω) can be obtained [18]:

$$\begin{cases} \nabla \cdot q + Q(\epsilon, d) \leq 0, & \text{in } B \\ q \cdot n_B \geq 0, & \text{on } B \end{cases} \quad (4)$$

where q is the flux of the phase field, $Q(\epsilon, d)$ is the phase field source, and n_B is the outer normal direction of the crack surface, which are respectively expressed as:

$$\begin{cases} q := \frac{2l}{c_a} G_f \nabla d \\ Q(\epsilon, d) = -\omega'(d) \bar{Y} - \frac{1}{c_a l} G_f \alpha'(d) \end{cases} \quad (5)$$

where G_f is the fracture energy of the solid material. ϵ is the solid structural strain; $\omega'(d)$ is the derivative of the energy degradation function. $\bar{Y}$ is the effective energy release rate. The energy degradation function $\omega(d)$ and the effective energy release rate $\bar{Y}$ are respectively defined as:

$$\begin{cases} \bar{Y} := \frac{\partial \psi}{\partial \omega} \\ \omega(d) = \frac{1}{1 + \phi(d)} \\ \phi(d) = \frac{a_1 d \cdot P(d)}{(1 - d)^r} \\ P(d) = 1 + a_2 \cdot d + a_3 \cdot d^2 \end{cases} \quad (6)$$

where ψ is the strain energy of the solid structure; r, a_1, a_2, a_3 are the coefficients of the degradation function, which can be determined based on experimental results.

At the same time, the displacement field ($\mathbf{u}$) of the solid structure satisfies the classic equilibrium equation:

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0, & \text{in } \Omega \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{t}, & \text{on } \partial\Omega \end{cases} \quad (7)$$

where $\boldsymbol{\sigma}$ is the stress tensor of the solid. $\mathbf{b}$ is the volume force on the structure. $\mathbf{n}$ is the outer normal direction of the solid structure. $\mathbf{t}$ is the force boundary condition on the structure.

2.2. Numerical implementation

Convert the governing equations of the phase field and displacement field into their weak forms:

$$\begin{cases} \int_{\Omega} \mathbf{b} \cdot \delta \mathbf{u} dV + \int_{\partial\Omega} \mathbf{t} \cdot \delta \mathbf{u} dA = \int_{\Omega} \boldsymbol{\sigma} : \nabla \delta \mathbf{u} dV \\ \int_B Q \delta d dV - \int_B \mathbf{q} \cdot \nabla \delta d dV \leq 0 \end{cases} \quad (8)$$

The finite element method is used for calculation. After discretizing the space, the displacement field, strain field, phase field and their gradients are respectively expressed as:

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &= \mathbf{N}a, \quad \boldsymbol{\varepsilon}(\mathbf{x}) = \mathbf{B}a \\ d(\mathbf{x}) &= \bar{\mathbf{N}}\bar{a}, \quad \nabla d = \bar{\mathbf{B}}\bar{a} \end{aligned} \quad (9)$$

where a is the displacement degree of freedom of the element node. $\mathbf{N}$ is the shape function of the displacement field. $\mathbf{B}$ is the displacement-strain coordination matrix of the displacement field. Correspondingly, $\bar{a}$ is the phase field degree of freedom of the element node. $\bar{\mathbf{N}}$ is the shape function of the phase field. $\bar{\mathbf{B}}$ is the phase field gradient coordination matrix. The specific expression can be referred to relevant finite element books [38].

From the weak form of the control equation, the equilibrium equation can be given as follows:

$$\begin{cases} \mathbf{r} = \mathbf{f}^{ext} - \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} dV = 0 \\ \bar{\mathbf{r}} = \int_B \bar{\mathbf{N}}^T Q dV - \int_B \bar{\mathbf{B}}^T \mathbf{q} dV \leq 0 \end{cases} \quad (10)$$

where $\mathbf{f}^{ext}$ is the external force borne by the structure.

The incremental iteration method is usually used to solve the above system of equations, that is, the time/load step $[0, T]$ is divided into M sub-steps, $m = 0, 1, \dots, M-1$. For the typical sub-step $\Delta t := t_{m+1} - t_m$ in the incremental step, all the state variables at time t_m are known, and on this basis, the state variables at time t_{m+1} are solved iteratively.

Considering the irreversibility of phase field evolution, the historical field variable H is introduced to replace the energy release rate $\bar{Y}$:

$$\begin{cases} H = \max_{m \in [0, T]} (\bar{Y}_0, \bar{Y}_{m+1}) \\ Q(\boldsymbol{\varepsilon}, d) = -\omega'(d)H - \frac{G_f}{c_d l} \alpha'(d) \end{cases} \quad (11)$$

where $\bar{Y}_0 = \frac{f_c^2}{2E_0}$ is the initial threshold of $\bar{Y}$. Cracks will appear if and only if condition $\bar{Y} > \bar{Y}_0$ is satisfied. Of course, due to the different mechanical behaviors of tension and compression, the energy release rate $\bar{Y}$ can be redefined, and the corresponding E_0 will change. For details, please refer to the corresponding research [18,39].

Then the inequality of the crack phase field sub-problem is transformed into the equation:

$$\begin{cases} \mathbf{r} = \mathbf{f}^{ext} - \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} dV = 0 \\ \bar{\mathbf{r}} = \int_B \bar{\mathbf{N}}^T Q dV - \int_B \bar{\mathbf{B}}^T \mathbf{q} dV = 0 \end{cases} \quad (12)$$

2.3. Program implementation steps

At the current stage, there are mainly two methods for solving phase field equations: the monolithic algorithm and the staggered iterative algorithm. The former is characterized by good convergence but requires a longer computation time. The latter, although it carries the risk of convergence difficulties, is easy to implement, robust, and efficient in computation, and thus is more frequently applied. Additionally, the use of the staggered iterative method facilitates the consideration of the effects of the initial state of quasi-brittle structures, which is why this method is adopted in this paper.

Based on the governing equations and the basic principles of finite elements, a phase field simulation method that considers the initial state of the solid structure can be obtained. Before applying an external load for crack propagation simulation, it is first

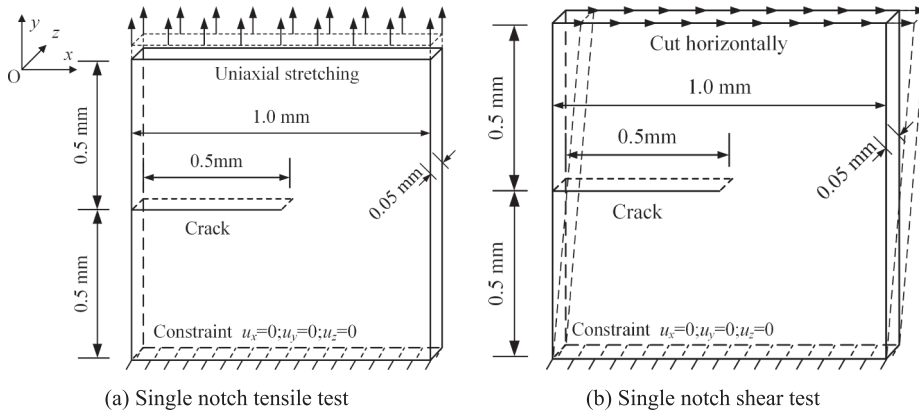


Fig. 1. Geometric and boundary conditions of mode I and mode II crack propagation.

Table 1

Calculation parameters of benchmark examples.

Parameter	Symbol	Value
Lamé's 1. parameter	λ	121.15 kN/mm ²
Lamé's 2. parameter	μ	80.77 kN/mm ²
Regularization width	l	0.015 mm
Griffith's constant	G_f	2.7 N/mm

Table 2

Parameters of AT2 model.

Model	c_a	a_1	a_2	a_3	r	ξ	f_i (Pa)
AT2	2.0	2.0	-0.5	0	2.0	0	0

necessary to go through an initial equilibrium iteration to obtain the initial stress value and phase field value of the structure for calculation. The implementation steps are as follows:

Step 1: The phase field degree of freedom $\bar{a} = \bar{a}_0$ is fixed, and the initial stress field is iterated. Apply the structure's initial stress (σ_0), strain (ε_0), displacement (u_0), etc. as loads to the structure, and add constraints at the same time. It should be noted that for all initial situations, in the first step of solution, the phase field and initial displacement degree of freedom can be assumed as $\bar{a}_0 = 0$ and $a_0 = 0$.

$$\left\{ \begin{array}{l} K_{dd} \delta a^1 = \int_{\Omega} B^T \omega(d_0) DB dV \delta a^1 = - \int_{\Omega} B^T (\sigma_0 + D \varepsilon_0 + DB u_0) dV = r \\ a_0^1 = a_0 + \delta a^1 \\ \sigma_0^1 = DB a_0^1 \end{array} \right. \quad (13)$$

where D is the elastic matrix. σ_0 is the initial stress. ε_0 is the initial strain. u_0 is the initial displacement. It should be noted that the general structure only needs to take one of the initial stress, initial strain, and initial displacement for estimation and calculation.

Step 2: The displacement field degree of freedom $a = a_0^1$ is fixed, and the initial phase field is iterated. Solve the following phase field equation to obtain the initial phase field under the initial stress load σ_0^1 .

$$\left\{ \begin{array}{l} K_{dd} \delta \bar{a}_0^1 = \int_B \bar{N}^T \left(-\frac{\partial Q}{\partial d} \right) \bar{N} + \frac{2l}{c_a} G_f \bar{B}^T \bar{B} dV \delta \bar{a}_0^1 = \int_B \bar{N}^T Q dV - \int_B \bar{B}^T q dV = \bar{r} \\ Q(\sigma_0, \bar{a}_0) = -\omega'(\bar{a}_0) H_1 - \frac{G_f}{c_a l} \alpha'(\bar{a}_0) \\ H_1 = \max \left(\frac{\max(f_i, [\sigma_0^1]^1)}{2E_0}, H_0 \right) \\ \bar{a}_0^1 = \bar{a}_0 + \delta \bar{a}_0^1 \end{array} \right. \quad (14)$$

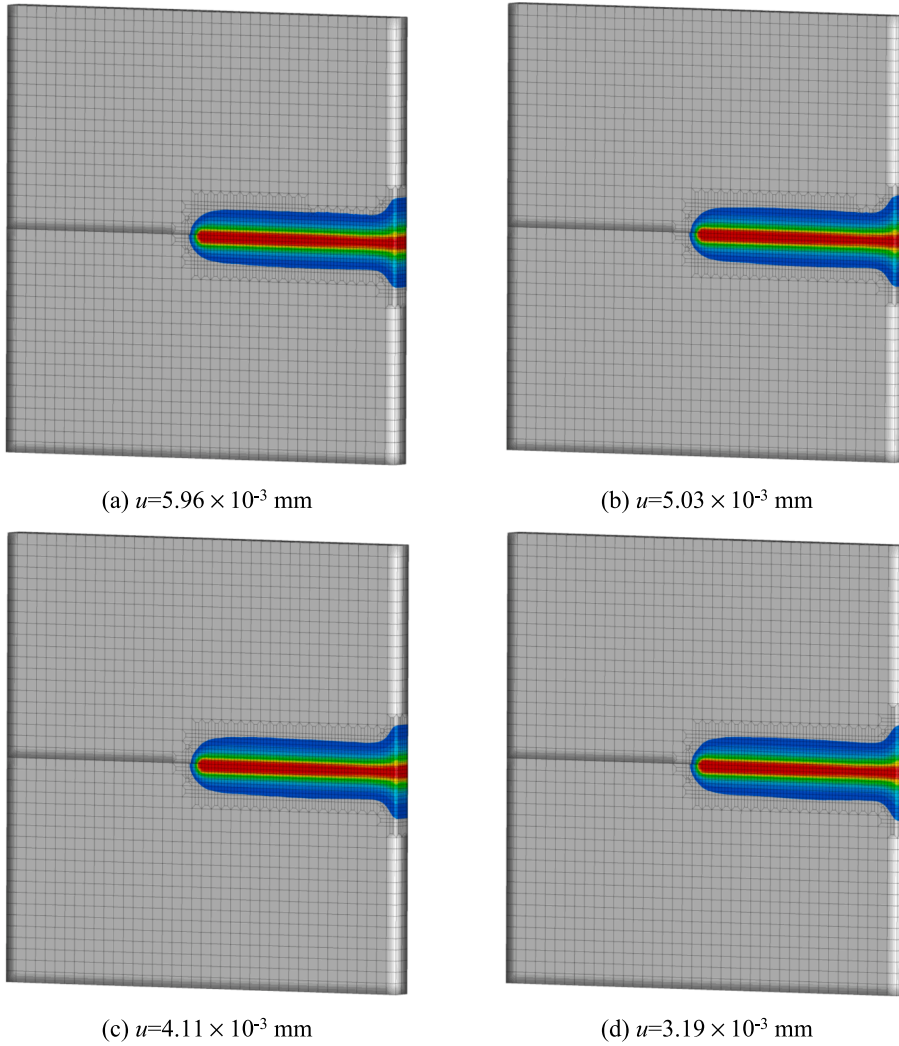


Fig. 2. Mode I crack propagation paths under different initial stresses (AT2 model).

where $[\sigma_0^1]^1$ is the first principal stress of σ_0^1 ; H_0 is the initial historical field variable, which can be set to 0.

Step 3: To judge the phase field value, if $\|\delta \bar{a}_0^1\|_2 < \zeta$ is satisfied, where $\|\cdot\|_2 := \sqrt{\sum_{i=1}^n (\cdot)^2}$ is the L_2 norm, and the tolerance ζ can be selected as $\zeta = 1.0 \times 10^{-5}$. Then take $(a_0^1, \bar{a}_0^1)$ as the initial stress field and phase field value. If $\|\delta \bar{a}_0^1\|_2 \geq \zeta$, replace $(a_0, \bar{a}_0)$ with $(a_0^1, \bar{a}_0^1)$, and repeat the step 1 and 2 until the phase field value converges. Finally, the calculated initial displacement is used to derive the initial stress σ_0 used for subsequent calculations.

The reason for carrying out the initial equilibrium iteration is that the estimated initial state of the structure in actual engineering is not completely accurate, and usually cannot meet the strain compatibility equation. If adopted directly without initial iteration, the calculation results may not converge. In the initial iteration, the initial stress, strain, and displacement are applied to the structure as loads, and the calculated stress, strain, and displacement satisfy the strain compatibility equation. In subsequent calculations, these calculation results can be used directly without computation non-convergence.

Afterwards, the load can be applied normally and the crack propagation calculation can be performed. For the initial iteration results, only the initial stress σ_0 and the initial phase field value $\bar{a}_0$ are retained, and the remaining variables are set to 0. The advantage of this is that all possible initial strains and displacements can be converted into initial stress conditions. At the same time, the improved sub-problem staggered iteration method takes the initial stress σ_0 as the stress load term into the calculation to solve the phase field equation. Taking a certain incremental step $[t_m, t_{m+1}]$ and the k -th iteration process as an example, the steps are as follows:

Step 1: Fixed phase field degree of freedom $\bar{a} = \bar{a}^{(k-1)}$, solve the following formula to obtain the nodal displacement a^k , and nodal stress σ^k :

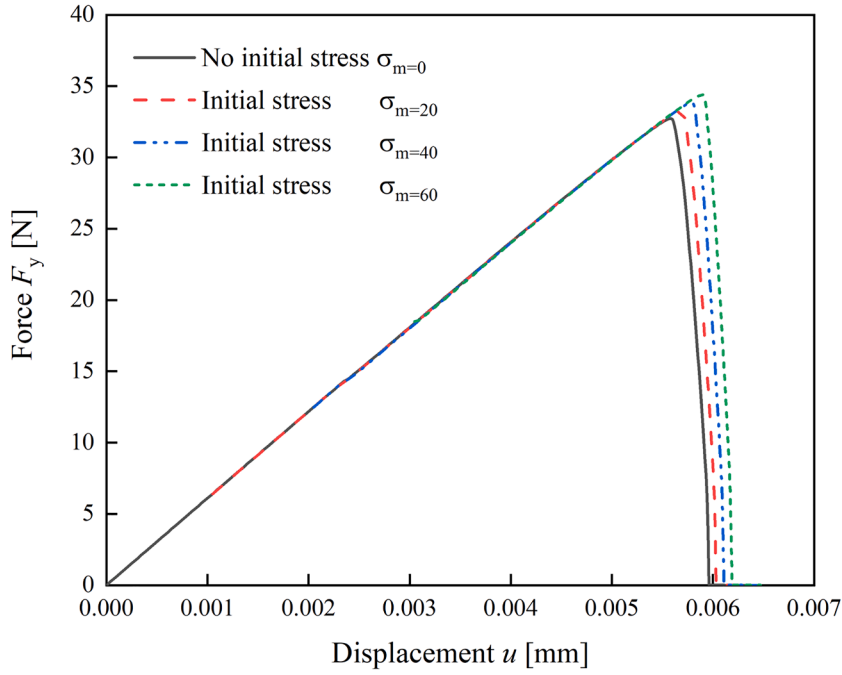


Fig. 3. Displacement-force curves of mode I crack propagation under different initial stresses (AT2 model).

$$\left\{ \begin{array}{l} K_{uu} \delta a^k = \underbrace{\int_{\Omega} B^T \omega(\bar{a}) DB dV}_{\bar{\sigma} = \omega(\bar{a}) \cdot (\sigma_0 + DB a^{k-1})} \delta a^k = f^{ext} - \underbrace{\int_{\Omega} B^T \bar{\sigma} dV}_{a^k = a^{k-1} + \delta a^k} = r \\ \sigma_k = \sigma_0 + DB a^k \end{array} \right. \quad (15)$$

Step 2: Fix the displacement field degree of freedom $a = a^k$, and solve the following formula to obtain the node phase field value $\bar{a}^k$.

$$\left\{ \begin{array}{l} K_{dd} \delta \bar{a}^k = \underbrace{\int_B \left[\bar{N}^T \left(-\frac{\partial Q}{\partial d} \right) \bar{N} + \frac{2l}{c_a} G_f \bar{B}^T \bar{B} \right] dV}_{Q(\sigma_k, \bar{a}^{(k-1)}) = -\omega'(\bar{a}^{(k-1)}) H^k - \frac{G_f}{c_a l} \alpha'(\bar{a}^{(k-1)})} \delta \bar{a}^k = \underbrace{\int_B \bar{N}^T Q dV - \int_B \bar{B}^T q dV}_{H^k = \max \left(\frac{\max(f_t, [\sigma_k]^1)}{2E_0}, H^{(k-1)} \right)} = \bar{r} \\ \bar{a}^k = \bar{a}^{(k-1)} + \delta \bar{a}^k \end{array} \right. \quad (16)$$

Step 3: To judge the phase field value, if $\|\delta \bar{a}_0^1\|_2 < \zeta$ is satisfied, the iteration is terminated and $(a^k, \bar{a}^k)$ is output as the displacement and phase field value under the current load. If it is not satisfied, repeat the step 1 and 2 until the calculation results converge.

In the previous phase field program calculation package (PFM) [34,35,40] developed based on MATLAB, the calculation content of the initial state of the structure was added, which can be obtained from the corresponding author.

3. Numerical examples and verification

3.1. Benchmark example test

3.1.1. Models and parameters

A classic example of a single-edge cutout thin plate is used to verify the phase field method (PFM) considering the initial state of the solid structure. As shown in Fig. 1, a horizontal crack with a length of 0.5 mm is prefabricated in the middle area of the left side. To facilitate research, the thickness of the thin plate is set to 0.05 mm. The mode I crack propagation under uniaxial tension (Fig. 1a) and the mode II crack propagation under horizontal shear (Fig. 1b) are calculated respectively.

The same calculation as in the previous study uses a homogenized eight-node hexahedral element with an element side length $h = 1/40$ mm. An adaptive refinement mesh is used in the calculation process to improve calculation efficiency.

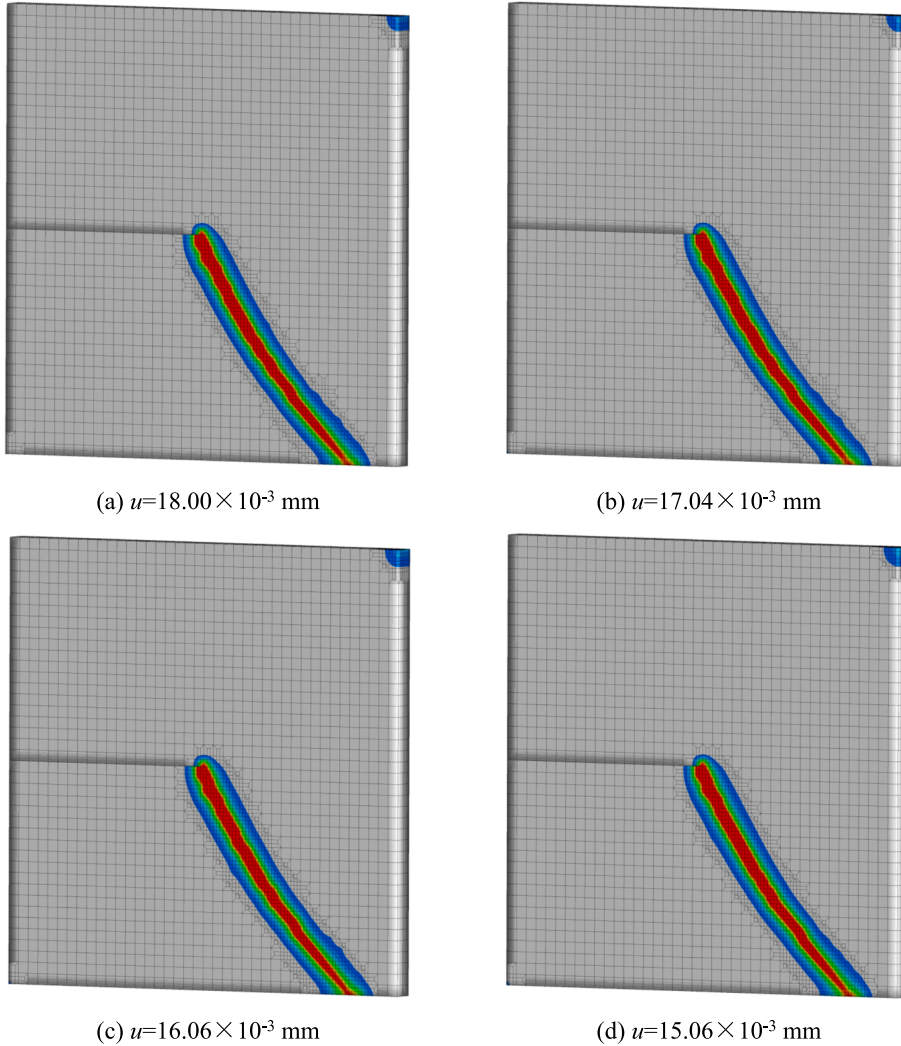


Fig. 4. Mode II crack propagation paths under different initial stresses (AT2 model).

Parameters required for calculation include Lamé's parameters (λ, μ), crack regularization width (l), Griffith's constant (G_f), etc. For convenience of comparison, the values of the calculation parameters in reference [7,41] are referenced, and the values of each parameter are shown in Table 1. First, the classic AT2 model is used for verification, which parameters c_a, r , etc. are shown in Table 2.

3.1.2. Uniaxial tensile test

The constraint and load of the thin plate under uniaxial tension are shown in Fig. 1a. The bottom of the model is a fixed as $u_x = 0$, $u_y = 0$, and $u_z = 0$. Four different initial stress states are compared. The first type has no initial stress and serves as the basic working condition. The top of the model is subjected to an upward displacement load in y-direction. The displacement increment of the first 100 steps is 5.0×10^{-5} mm, the next steps are 1.0×10^{-5} mm. The second type has initial stress, and its initial stress value (σ_0) is equal to the stress value output when the calculation reaches 20 steps in the first condition ($\sigma_{m=20}$). The third type initial stress value (σ_0) is equal to the stress value output when the calculation reaches 40 steps in the first condition ($\sigma_{m=40}$). The fourth type initial stress value (σ_0) is equal to the stress value output when the calculation reaches 60 steps in the first condition ($\sigma_{m=60}$).

Fig. 2 shows the final mode I crack propagation path calculated using different initial stresses. It can be seen from the figure that under different initial stresses, cracks will expand after the structure is subjected to different displacement loads. Moreover, the crack expansion paths all penetrate the thin plate along the horizontal direction, which is the same as the results simulated by previous studies [42,43]. During the crack propagation process, the range of $d \geq 0.1$ is also the same. When there is initial stress in the structure, the adaptive refinement mesh proposed in previous research [35] can still work well, and the elements on the crack expansion path are automatically refined.

Fig. 3 shows the displacement-force curve in the y direction at the top of the model when the mode I crack propagation under different initial stresses. It should be noted that for the convenience of comparison, we moved the displacement-force curves of

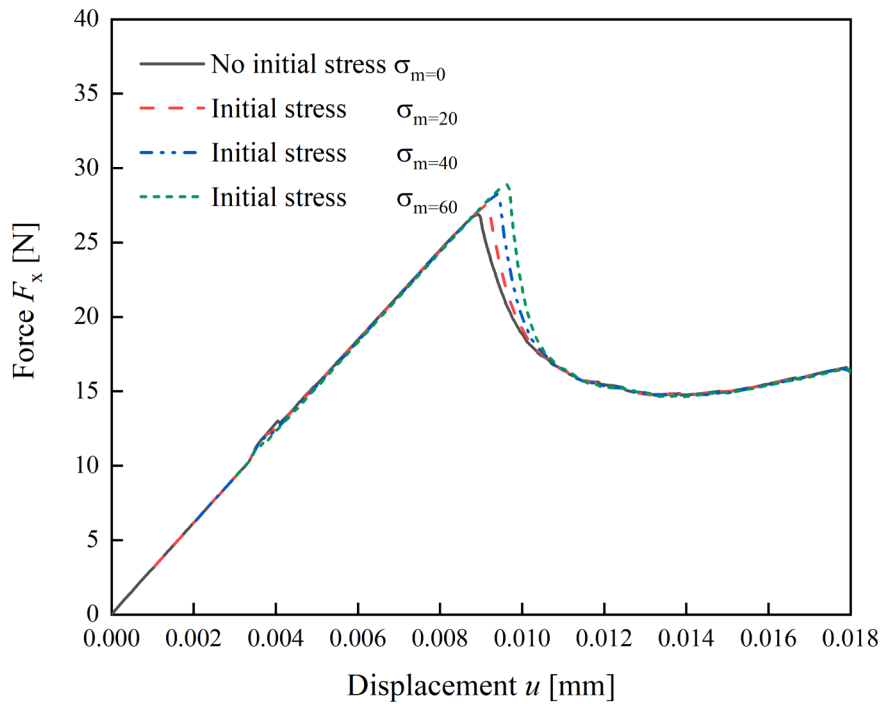


Fig. 5. Displacement-force curves of mode II crack propagation under different initial stresses (AT2 model).

Table 3

Parameters of PF_CZM model.

Model	c_a	a_1	a_2	a_3	r	ξ	$f_i(\text{Pa})$
PF_CZM	π	$\frac{4G_f \cdot E_0}{\pi l \cdot f_t^2}$	1.3868	0.6567	2.0	2.0	3.0×10^9

conditions (2), 3, and 4 by 20, 40, and 60 load steps respectively along the x direction of the coordinate axis. It can be seen from Fig. 3 that in the stage when the crack does not expand, the calculated displacement-force curves are the same. However, as the initial stress of the structure increases, the calculated crack initiation displacement and peak force will also increase. Under the condition of no initial stress, when the displacement load reaches 5.60×10^{-3} mm, the peak load is 32.75 N. When the initial stress is $\sigma_{m=60}$ and the displacement load reaches 5.85×10^{-3} mm, the peak load is 34.35 N. The calculated peak load difference is 1.60 N, and the calculation error is 4.89 %. The calculated crack initiation displacement difference is 0.25×10^{-3} mm, and the calculation error is 4.46 %.

3.1.3. Horizontal shear test

The constraint and load of the thin plate under horizontal shear are shown in Fig. 1b. The bottom of the model is also fixed as $u_x = 0$, $u_y = 0$, and $u_z = 0$. A horizontal rightward displacement load is applied in the x direction of the top of the model. Similar to uniaxial tensile test, four different initial stress states are still compared. The first type has no initial stress and serves as the basic working condition. The displacement increment of the first 100 steps is 5.0×10^{-5} mm, the next steps are 2.0×10^{-5} mm. The same as the uniaxial tensile test, the initial stress in the second, third, and fourth calculation conditions is equal to the stress value output when the calculation reaches 20 ($\sigma_{m=20}$), 40 ($\sigma_{m=40}$), and 60 ($\sigma_{m=60}$) steps in the first condition, respectively.

Fig. 4 shows the final mode II crack propagation path calculated using different initial stresses. It can be seen from the figure that the simulated crack growth paths are the same in the four conditions. The cracks all expanded diagonally downward on the right side, and the path showed a certain degree of curvature, which was the same as the results simulated by previous studies [42,43]. Similarly, in this test, the adaptive refinement mesh can still automatically refine the elements on the crack expansion path.

Fig. 5 shows the displacement-force curve in the x direction at the top of the model when the mode II crack propagation under different initial stresses. The same as the uniaxial tensile test, the displacement-force curve of conditions (2), 3, and 4 moved by 20, 40, and 60 load steps respectively along the x direction of the coordinate axis. As can be seen from Fig. 5, the calculated displacement-force curves are the same as those for mode I in the non-crack expansion stage. And as the initial stress increases, the calculated crack initiation displacement and peak load will also increase. Under the condition of no initial stress, when the displacement load reaches 8.96×10^{-3} mm, the peak load is 26.92 N. When the initial stress is $\sigma_{m=60}$ and the displacement load reaches 9.58×10^{-3} mm, the peak load is 28.60 N. The calculated peak load difference is 1.68 N, and the calculation error is 6.24 %. The calculated crack initiation

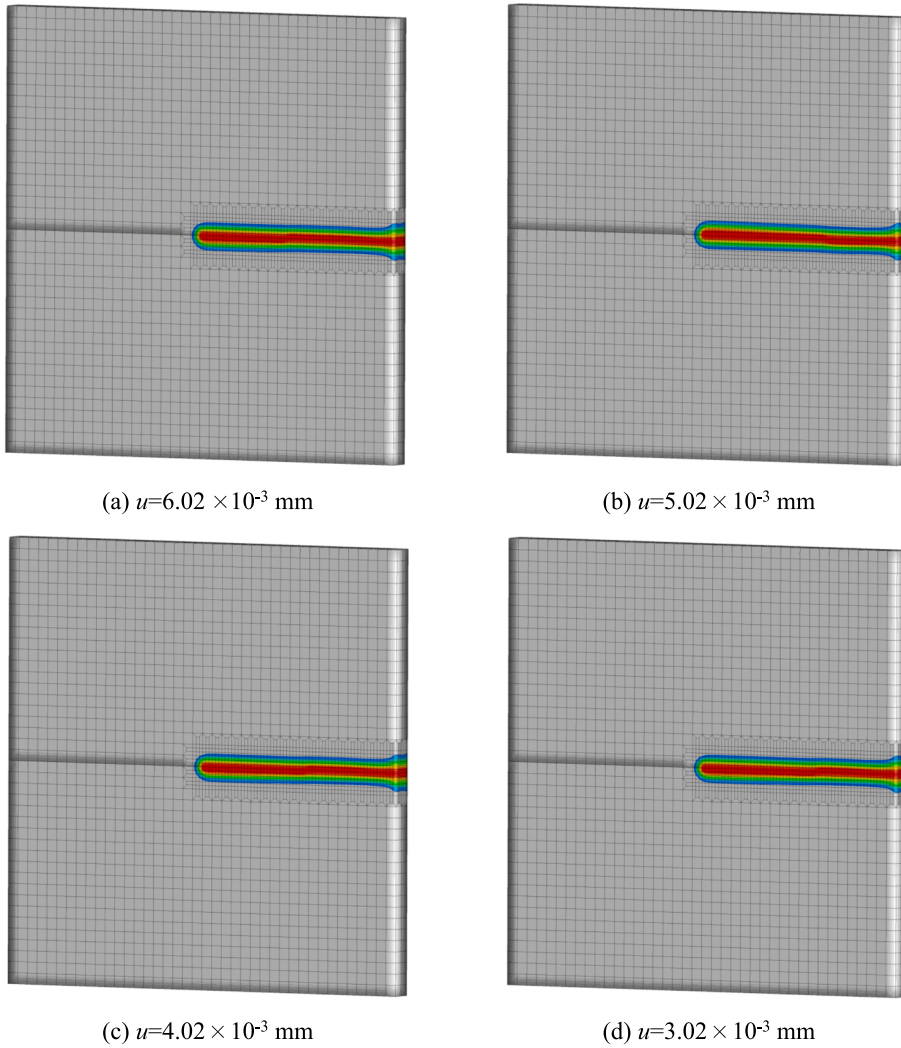


Fig. 6. Mode I crack propagation paths under different initial stresses (PF_CZM model).

displacement difference is 0.62×10^{-3} mm, and the calculation error is 6.92 %.

3.1.4. Different phase field models

We will still use different phase field models in engineering, such as the AT1 model for underground structural structures and the PF_CZM model for concrete structures. The AT1 model and the PF_CZM model have similar properties, that is, all areas of the structure are elastic before damage or crack initiation, with the main difference being the softening stage after crack initiation. Considering the brittle/quasi-brittle structure, no damage or cracks occur in the structure under the initial stress state. The PF_CZM model is used to calculate the uniaxial tensile and horizontal shear damage of the same thin plate to verify the applicability of the PFM to different phase field models. The calculation parameters of the PF_CZM model are shown in Table 3.

Under four different initial stresses, the final expansion morphology of mode I crack simulated by the PF_CZM model is shown in Fig. 6. It can be found that the same as the AT2 model, the cracks expand along the horizontal direction and penetrate the thin plate. The difference is that the area of $d \geq 0.1$ in the PF_CZM model is smaller. This is because the fracture geometry function used in the PF_CZM model determines its final phase field distribution as a finite fracture zone, while the AT2 model has an infinite fracture zone, so there is a difference.

Fig. 7 shows the displacement-force curve in the y direction at the top of the model when the mode I crack propagation under different initial stresses with PF_CZM model. As before, the displacement-force curve when the initial stress is applied is moved along the x direction of the coordinate axis by 20, 40, and 60 load steps for easy comparison. It can be seen from the figure that the displacement-force curves almost completely overlap under different initial stresses. The crack initiation displacement is all 5.87×10^{-3} mm, the peak load is 35.74 N, and the calculation error is less than 1 %.

Fig. 8 shows the final mode II crack propagation path calculated using different initial stresses with PF_CZM model. The crack

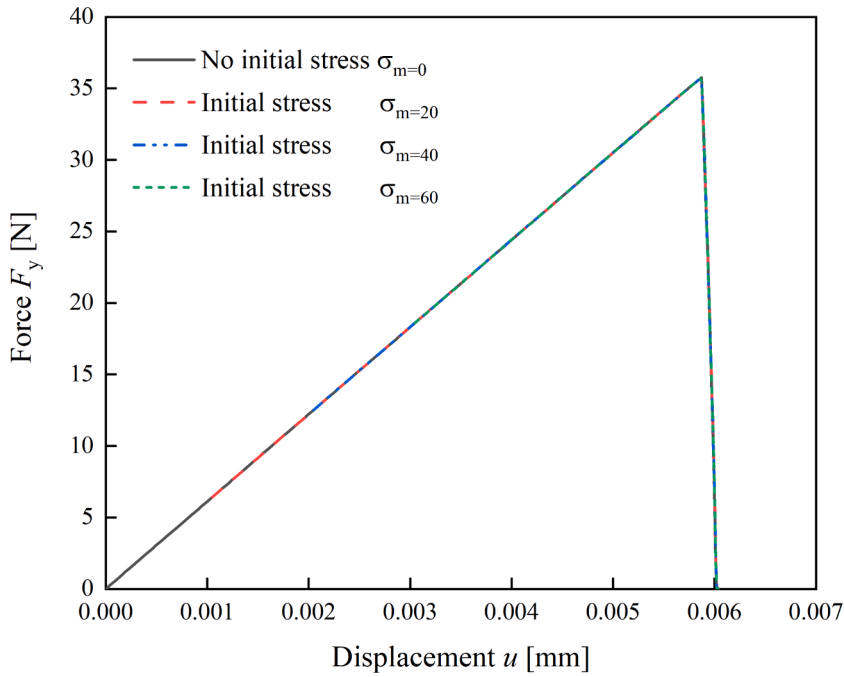


Fig. 7. Displacement-force curves of mode I crack propagation under different initial stresses (PF_CZM model).

expansion path is roughly the same as that simulated by the AT2 model, but the width of the crack zone is smaller. At the same time, it can be seen from Figs. 6 and 8 that the elements are automatically refined along the crack expansion path. This means that the adaptive refinement mesh method previously proposed based on AT2 model [35] is universal and is also applicable to other phase field models such as PF_CZM.

Fig. 9 shows the displacement-force curve in the x direction at the top of the model when the mode II crack propagation under different initial stresses with PF_CZM model. As before, the displacement-force curve when the initial stress is applied is moved along the x direction of the coordinate axis by 20, 40, and 60 load steps for easy comparison. Same as mode I, the displacement-force curves completely overlap under different initial stresses. The crack initiation displacement is all 9.34×10^{-3} mm, the peak load is 28.46 N, and the calculation error is less than 1 %. Based on Figs. 7 and 9, it can be concluded that using the phase field model proposed in this paper that considers the initial state of the structure, if similar phase field models such as PF_CZM model is used, accurate simulation results can be obtained.

When comparing the computational results of the AT2 model with those of the PF_CZM model, it is found that the results from the PF_CZM model are superior. This outcome is primarily due to the nature of the AT2 model, which lacks an elastic region before the onset of damage and provides non-zero damage values during the loading steps. Consequently, when the AT2 model is used, different initial stress values correspond to different non-zero phase field values. The phase field values obtained after initial iterations using the method described in this paper exhibit deviations from the actual phase field values. Furthermore, these deviations increase as the initial stress increases, leading to discrepancies in the calculated peak displacements and loads. On the other hand, models like PF_CZM, which have an elastic region before damage initiation, start with zero damage values, resulting in unbiased phase field values after initial iterations, and thus smaller discrepancies in the calculated peak displacements and loads.

Additionally, it must be highlighted that the use of an adaptive refinement strategy also affects the computational results. During the phase before crack propagation, calculations are conducted on a coarse mesh, and the initial equilibrium iterations are also performed on this coarse mesh. As a result, the phase field values obtained during this stage are influenced by the mesh size, and deviations in the phase field values are observed after iterations.

3.2. Uniform initial stress of cracked structure

It is difficult to accurately predict the initial stress in engineering, and often only approximate estimates can be given. In this section, the same model will be used, changing the initial stress to a uniform value, to verify the engineering applicability of the PFM. Taking the mode I crack propagation as an example, the PF_CZM model is used, and the no-initial stress state is used as the basic comparison working condition. Uniform initial stresses of -500 MPa, -200 MPa, 200 MPa, and 500 MPa are applied in the y direction of all nodes of the thin plate respectively, and the crack propagation of the structure under uniaxial tensile load is calculated.

According to the implementation steps, the initial iteration of the displacement and phase field is first carried out, taking the initial stress of -200 MPa as an example. Fig. 10a shows the initial estimated structural stress, and the stress field after the initial equilibrium iteration is shown in Fig. 10b. It can be found that the stress is redistributed after the equilibrium iteration, and the compressive stress

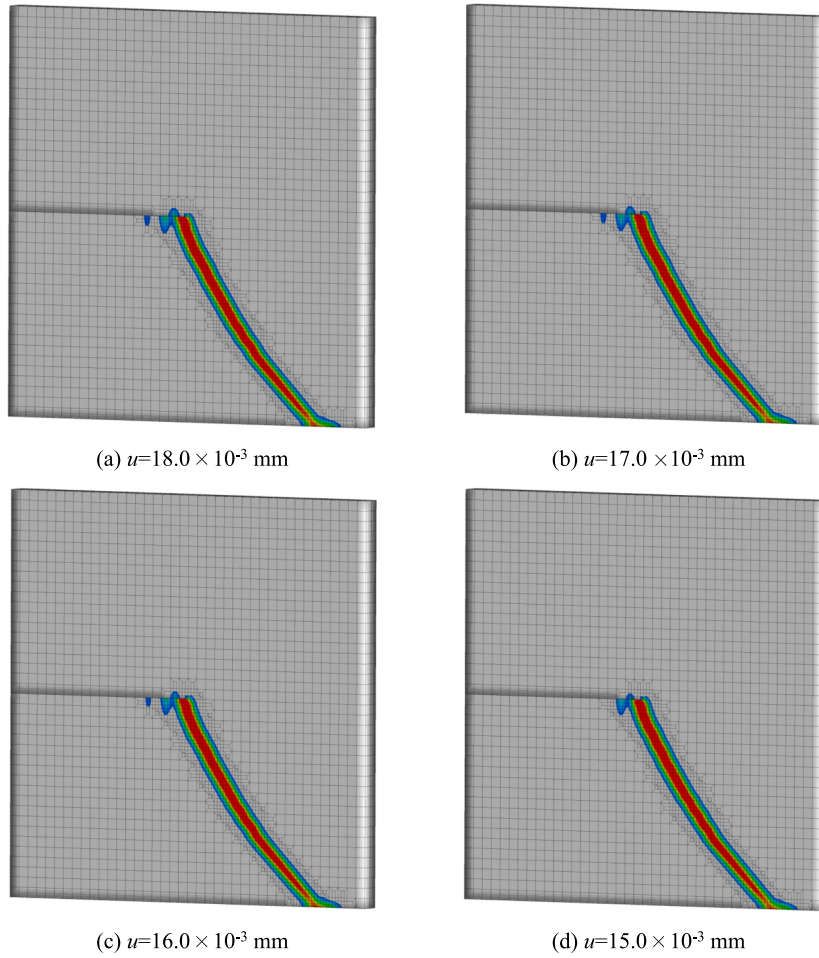


Fig. 8. Mode II crack propagation paths under different initial stresses (PF_CZM model).

at the tip increases significantly, reaching -700 MPa. This is because the estimated initial stress is relatively rough, and the impact of the presence of cracks on the stress cannot be carefully considered. But in fact, the existence of cracks will cause local stress concentration and increase the tip stress. Without an initial equilibrium iteration of the stress field, subsequent calculations will not converge.

Finally, the mode I crack expansion morphology under different uniform initial stresses is obtained, as shown in Fig. 11. It can be seen from the figure that the crack propagation path does not change significantly and remains consistent. However, the displacement load when the crack propagation is completed will change significantly. As the tensile stress in the direction gradually increases, the displacement load at fracture gradually decreases. At the same time, in this case, the adaptive refinement mesh can also work normally and automatically refine the mesh on the crack expansion path.

Fig. 12 shows the displacement-force curve in the y direction at the top of the model when the mode I crack propagation under different uniform initial stresses with PF_CZM model. It can be seen from the figure that as the initial tensile stress in the y direction increases, the crack initiation displacement of the structure continues to decrease, and there is an obvious linear relationship between the two. However, the peak load when the structure fails is the same, 35.74 N. The simulation results are consistent with the actual situation. This shows that the PFM considering the initial state of the structure has strong engineering applicability. Based on the stress data measured during the project, the initial stress field distribution of the structure is estimated. In engineering, the initial stress field distribution of the structure is estimated based on the measured stress data. Using the estimated initial stress field as input data, the PFM can calculate a more reasonable redistributed initial stress based on the structural characteristics and simulate the damage of the structure under corresponding loads.

4. Discussion

In engineering, we may also encounter situations where the initial strain or initial displacement of the structure is known. The phase field method (PFM) proposed in this paper can also handle these two special situations.

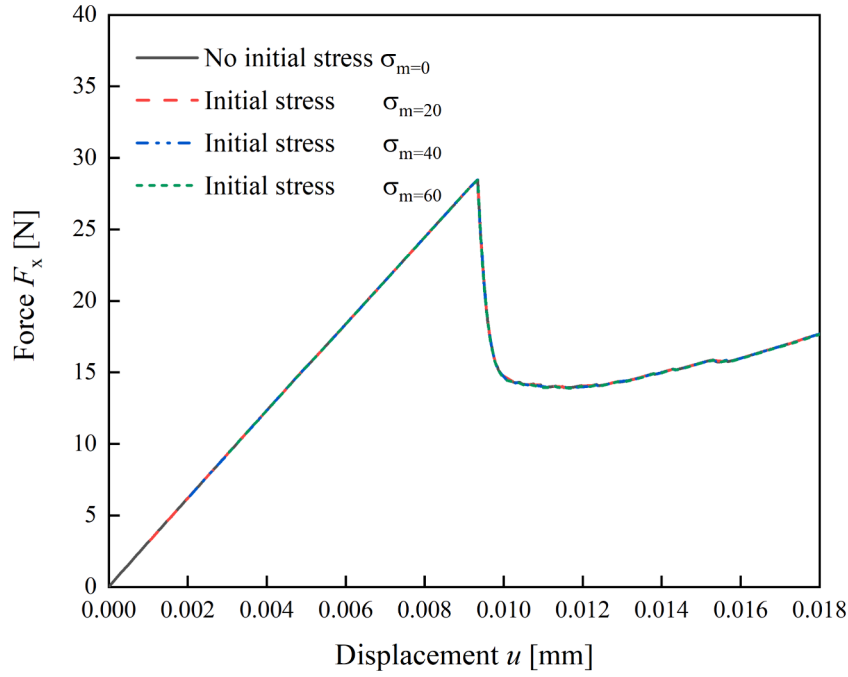


Fig. 9. Displacement-force curves of mode II crack propagation under different initial stresses (PF_CZM model).

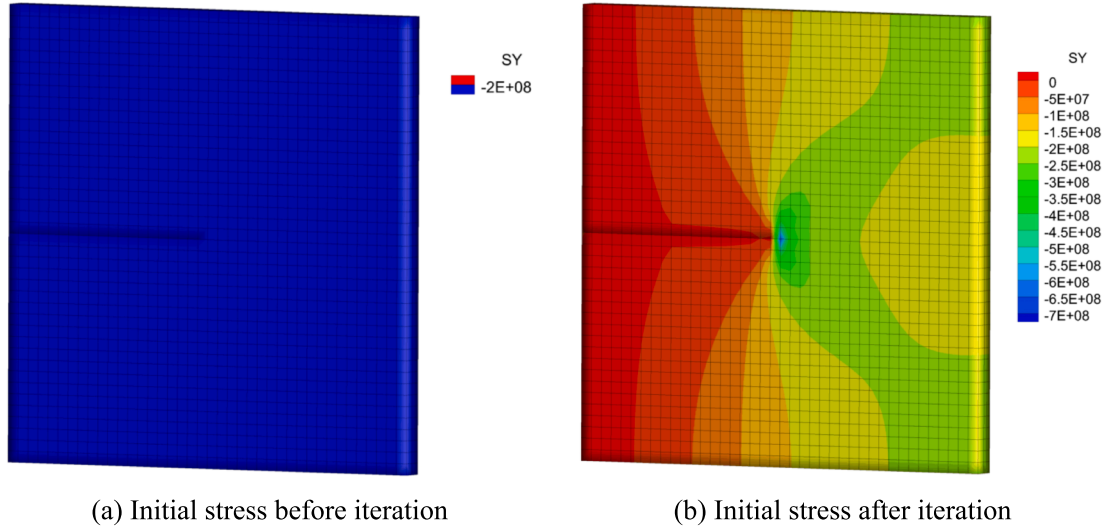


Fig. 10. Effects of initial equilibrium iterations.

4.1. Known initial strain

The basic working condition of uniaxial tension is still used, and the initial quantity is changed to strain. The same four calculation conditions are: without initial strain ($\epsilon_m=0$), with initial strain ($\epsilon_m=20$, $\epsilon_m=40$, $\epsilon_m=60$). The calculation uses the AT2 model. Since the calculated final crack propagation is the same, it will not be shown again. Fig. 13 shows the displacement-force curve in the y direction at the top of the model when the mode I crack propagation. Comparing Fig. 3 and Fig. 13, we can find that the two are almost the same. As the initial strain increases, the calculation error will gradually increase, but it still meets the requirements of engineering calculations.

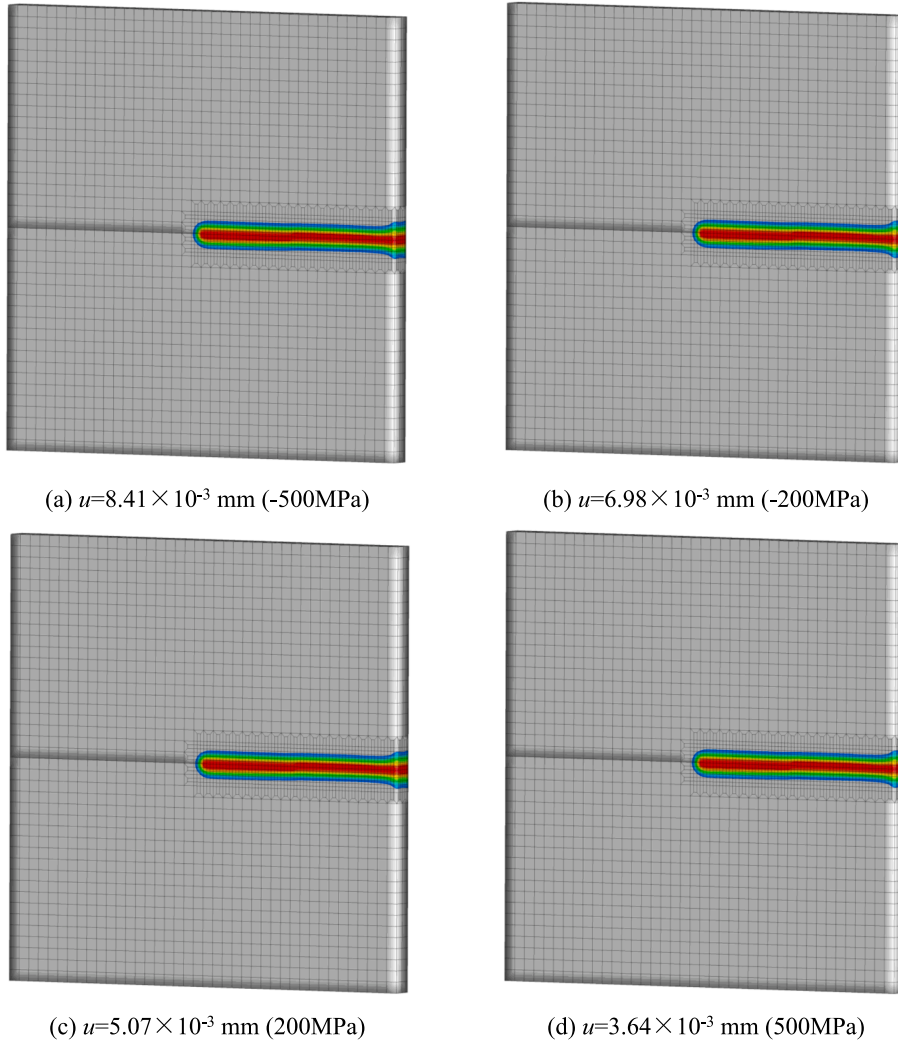


Fig. 11. Mode I crack propagation paths under different uniform initial stresses (PF_CZM model).

4.2. Known initial displacement

The basic working condition of uniaxial tension is still used, and the initial quantity is changed to displacement. The same four calculation conditions are: without initial displacement ($u_m=0$), with initial displacement ($u_m=20$, $u_m=40$, $u_m=60$). The calculation also uses the AT2 model. For the same reason, the crack propagation morphology will no longer be displayed. Fig. 14 shows the displacement-force curve in the y direction at the top of the model when the mode I crack propagation. The same as Figs. 3 and 13, as the initial displacement increases, the calculation error will gradually increase. Although the relative error calculated by the initial displacement is small, the calculation results of the three are still similar. The reason for this phenomenon is that the initial stress, initial strain, and initial displacement are all treated as the initial load of the structure during processing, and $\sigma_0 = D\epsilon_0 = BDu_0$ is basically satisfied for the stressed structure. In the calculation, the initial strain and displacement will be converted into initial stress for processing, so excluding any errors that may occur during the calculation process, the calculation results of the three are similar.

4.3. Rebar rust expansion test

The hydration reaction of cement releases a significant amount of heat, leading to an increase in the temperature of concrete. When the volume of a concrete structure is large, the dissipation of heat becomes challenging, causing the temperature in the central region of the structure to rise, while the temperature in the boundary areas decreases, creating a large temperature gradient. When the thermal deformation of concrete is constrained, such as by the foundation or the underlying older concrete, substantial thermal stresses are generated. These residual thermal stresses will remain throughout the operational phase of the structure [28]. Referring on previous studies [44], a model of single rebar rust expansion was established, and the cracking of the concrete protective layer caused by

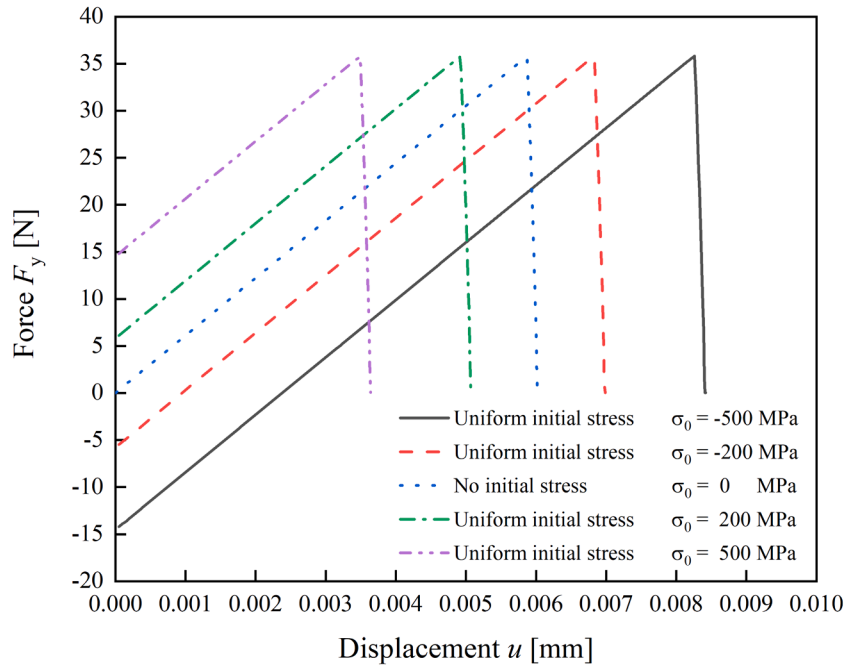


Fig. 12. Displacement-force curves of mode I crack propagation under different uniform initial stresses (PF_CZM model).

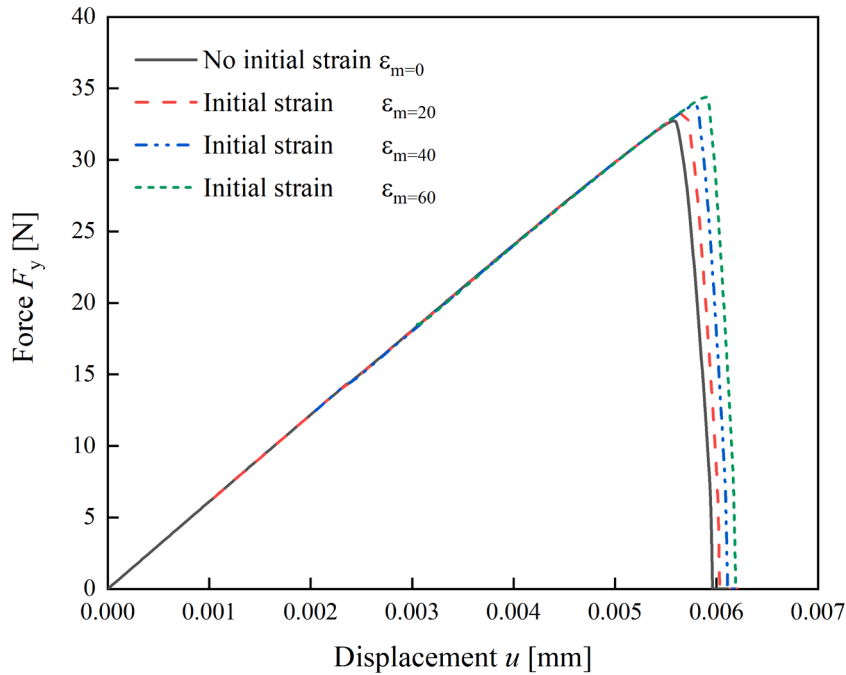


Fig. 13. Displacement-force curves of mode I crack propagation under different initial strain (AT2 model).

rebar rust expansion under two different cases was simulated. As shown in Figs. 15 and 16, the model dimensions are 200 mm $\times$ 200 mm $\times$ 300 mm, with a rebar diameter of 20 mm, a protective layer thickness of 50 mm, and a longitudinal length of the rebar being 300 mm. The rebar adopted a non-uniform rust expansion pattern. Simulations were carried out both with and without initial thermal stress. The rebar rust expansion function, material parameters, and PF_CZM model parameters used in the calculations are the same as those used in previous studies [44].

Fig. 17 illustrates the crack propagation when there is no initial stress, with the crack morphology largely resembling that simulated

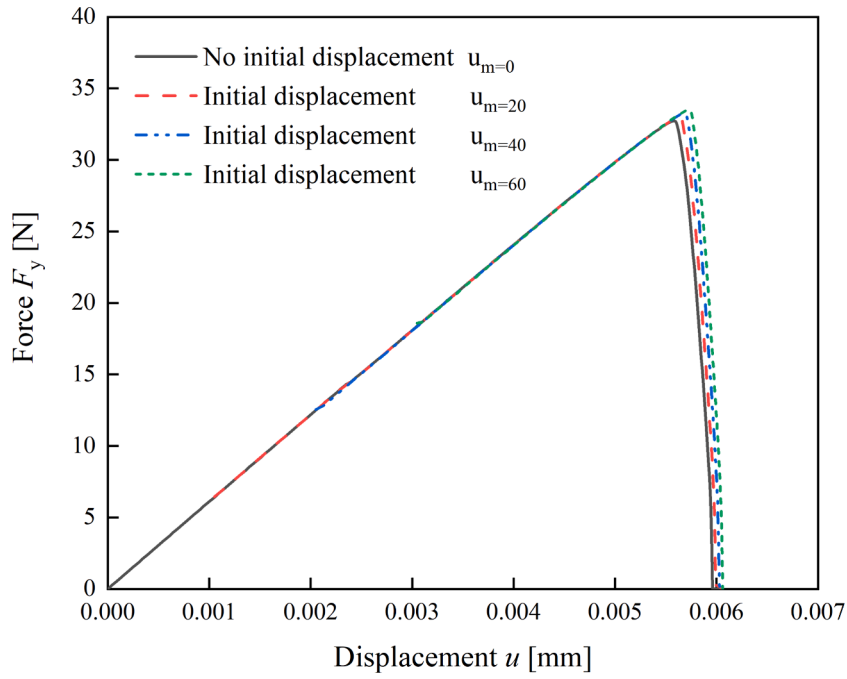


Fig. 14. Displacement-force curves of mode I crack propagation under different initial displacement (AT2 model).

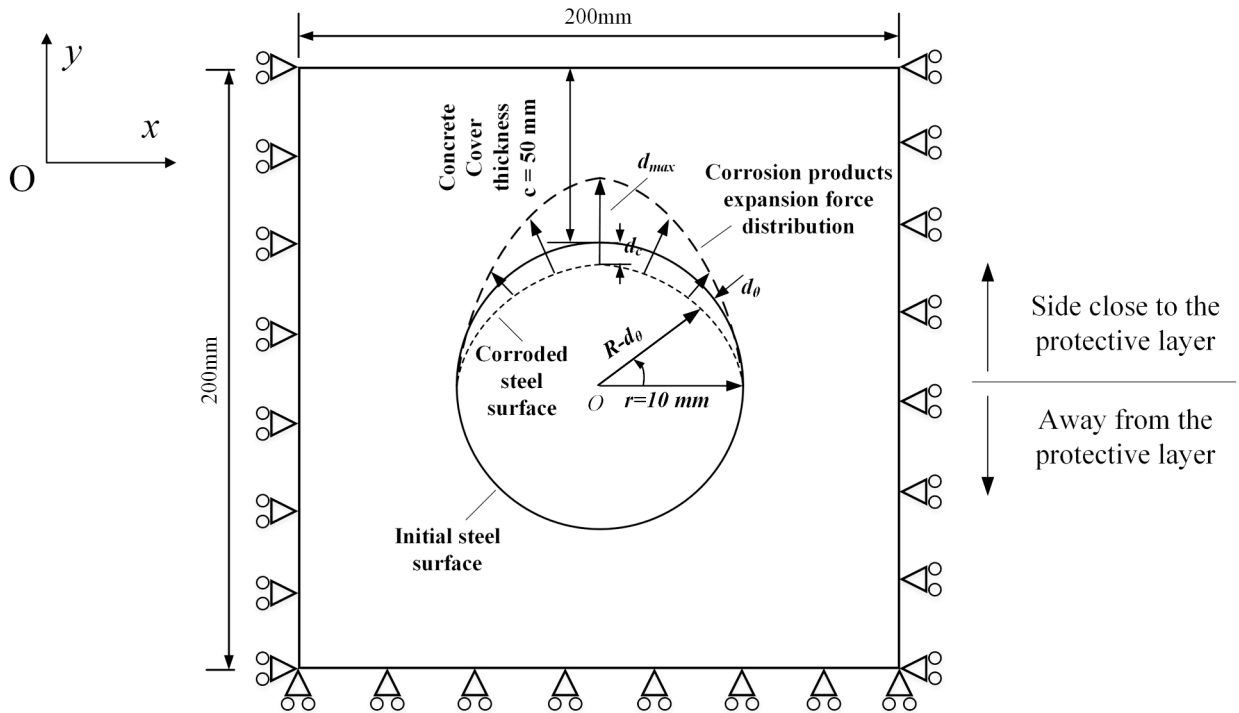


Fig. 15. Schematic diagram of corrosion expansion model of the rebar.

in the literature [44], where wedge-shaped damage occurs in the protective layer. A tensile stress of 2.0 MPa is applied in the model x direction to simulate the structure containing residual thermal stress. Fig. 18 shows the crack propagation under initial stress conditions. It can be observed that the crack morphology has significantly changed, with a vertical crack extending perpendicularly to the surface of the structure. Additionally, the angle between the two inclined cracks is noticeably reduced compared to the first case. More

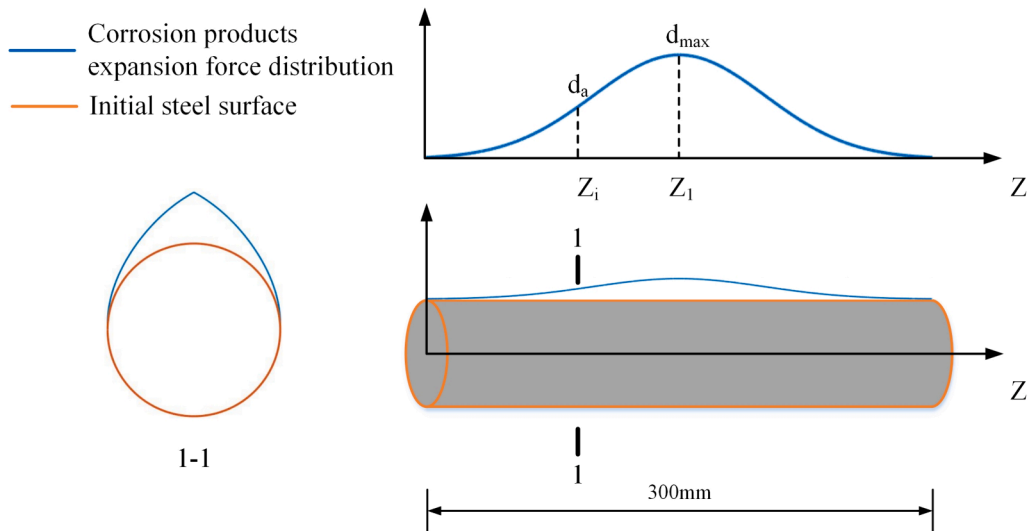


Fig. 16. schematic diagram of three-dimensional non-uniform corrosion morphology.

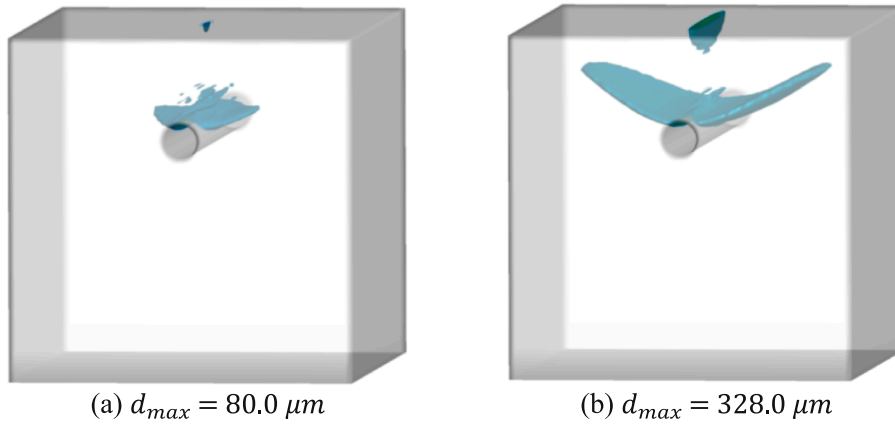


Fig. 17. Crack propagation without initial stress and initial damage under non-uniform corrosion ($\alpha = 4$ c = 20 mm).

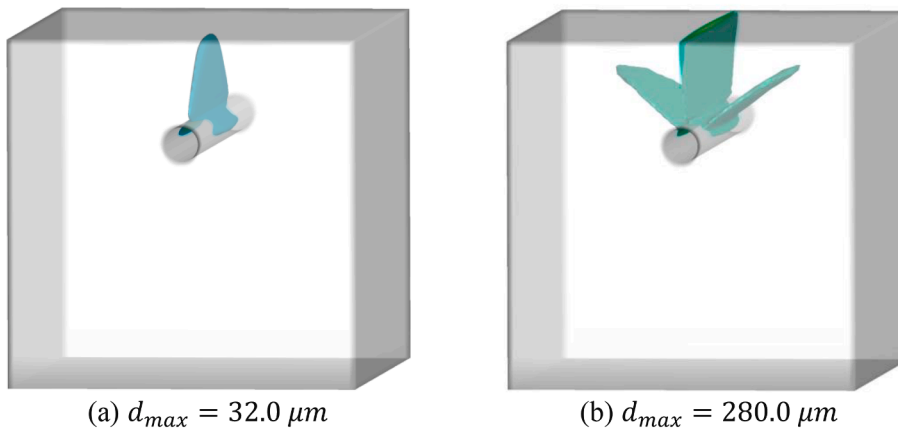


Fig. 18. Crack propagation with initial stress ($\sigma_x = 2.0$ MPa) and without initial damage under non-uniform corrosion ($\alpha = 4$ c = 20 mm).

importantly, compared to the first case, the maximum corrosion expansion displacement of the rebar when significant damage occurs in the concrete is reduced from 80.0 μm to 32.0 μm , and the maximum corrosion expansion displacement of the rebar when significant cracks appear is reduced from 328.0 μm to 280.0 μm . This implies that the residual tensile stress in the structure will reduce its durability, which is consistent with practical observations. This example also demonstrates the engineering applicability of the model discussed in this paper to a certain extent.

Although the PFM considering the initial state of quasi-brittle structures proposed in this paper has certain applicability, it has not been focused on the crack propagation in elastoplastic metallic materials or viscoelastic rubber-like materials. Therefore, it cannot be inferred that this model is suitable for the calculation of crack propagation in arbitrary materials. In the next phase of research, attention will be given to the fracture simulation of elastoplastic metals such as reinforced steel, to validate and improve the model so that it can be adapted for the fracture simulation of elastoplastic materials.

5. Conclusion

In some actual engineering simulations, the structure already contains initial stresses before external loads are applied. The initial state of the structure will inevitably affect the initiation and expansion of cracks when it is damaged. To address the simulation problem of crack propagation in solid structures with initial stress states, this paper proposes a phase-field method (PFM) that considers the initial state of quasi-brittle structures based on the fundamental principles of the method. Through initial equilibrium iteration, the estimated initial stress, initial strain and initial displacement in the structure are converted into the initial load of the structure, and the redistributed initial stress of the structure is obtained. The redistributed stress is then used as the internal initial stress load term of the structure to participate in the calculation, and an improved sub-problem staggered iteration method is used to simulate the crack growth of the structure under external loads.

The AT2 and PF_CZM models were used respectively to simulate the crack growth of prefabricated notched thin plates in uniaxial tension and horizontal shear under different initial stresses, initial strains and initial displacements. Research shows that for different phase field models, the PFM can better simulate the crack propagation of the quasi-brittle structure with the initial state. The calculation error for the PF_CZM model is within 1 %, and the calculation error for the AT2 model increases with the increase of initial stress, but is less than 7 %. In addition, it is also verified that the adaptive refinement mesh proposed based on the AT2 model is universal to other phase field models.

At the same time, the crack propagation problem of cracked structures under different uniform initial stresses is accurately simulated. It shows that the initial equilibrium iteration can reasonably redistribute the estimated initial stress according to the structural characteristics, thereby solving the problem of calculation non-convergence caused by inaccurate estimation of the structure initial state in actual engineering. It reflects the applicability of this method to actual engineering and provides a reference for solving practical engineering problems.

CRedit authorship contribution statement

Caihong Zhang: Writing – review & editing, Writing – original draft, Software, Conceptualization. **Jiaming Zhang:** Methodology, Software, Writing – review & editing. **Hanzhang Li:** Software, Investigation. **Yu Li:** Writing – original draft, Methodology, Investigation. **Wenqiang Xu:** Writing – original draft, Software, Methodology, Conceptualization. **Sheng Qiang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Min Yuan:** Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (No. 52079049), Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX22_0641), and Ningbo City Water Conservancy Science and Technology Project (NSKA202343).

References

- [1] Johnson GR, Stryk RA. Eroding interface and improved tetrahedral element algorithms for high-velocity impact computations in three dimensions. *Int J Impact Eng* 1987;5:411–21.
- [2] Xu X-P, Needleman A. Numerical simulations of fast crack growth in brittle solids. *J Mech Phys Solids* 1994;42:1397–434.
- [3] Song J-H, Wang H, Belytschko T. A comparative study on finite element methods for dynamic fracture. *Comput Mech* 2008;42:239–50.
- [4] Sam C-H, Papoulia KD, Vavasis SA. Obtaining initially rigid cohesive finite element models that are temporally convergent. *Eng Fract Mech* 2005;72:2247–67.

- [5] Belytschko T, Black T. Elastic crack growth in finite elements with minimal remeshing. *Int J Numer Meth Eng* 1999;45:601–20.
- [6] Moës N, Dolbow J, Belytschko T. A finite element method for crack growth without remeshing. *Int J Numer Meth Eng* 1999;46:131–50.
- [7] Miehe C, Hofacker M, Welschinger F. A phase field model for rate-independent crack propagation: Robust algorithmic implementation based on operator splits. *Comput Methods Appl Mech Eng* 2010;199:2765–78.
- [8] Francfort GA, Marigo J-J. Revisiting brittle fracture as an energy minimization problem. *J Mech Phys Solids* 1998;46:1319–42.
- [9] Gravouil A, Moës N, Belytschko T. Non-planar 3D crack growth by the extended finite element and level sets—Part II: Level set update. *Int J Numer Meth Eng* 2002;53:2569–86.
- [10] Sukumar N, Chopp DL, B  chet E, Mo  s N. Three-dimensional non-planar crack growth by a coupled extended finite element and fast marching method. *Int J Numer Meth Eng* 2008;76:727–48.
- [11] Mumford DB, Shah J. Optimal approximations by piecewise smooth functions and associated variational problems. *Commun Pure Appl Math* 1989.
- [12] Tann   E, Li T, Bourdin B, Marigo J-J, Maurini C. Crack nucleation in variational phase-field models of brittle fracture. *J Mech Phys Solids* 2018;110:80–99.
- [13] Miehe C, Welschinger F, Hofacker M. Thermodynamically consistent phase-field models of fracture: Variational principles and multi-field FE implementations. *Int J Numer Meth Eng* 2010;83:1273–311.
- [14] Wu J-Y, Xu S-L. Reconsideration on the elastic damage/degradation theory for the modeling of microcrack closure-reopening (MCR) effects. *Int J Solids Struct* 2013;50:795–805.
- [15] Mesgarnejad A, Bourdin B, Khonsari M. Validation simulations for the variational approach to fracture. *Comput Methods Appl Mech Eng* 2015;290:420–37.
- [16] Pham K, Amor H, Marigo J-J, Maurini C. Gradient damage models and their use to approximate brittle fracture. *Int J Damage Mech* 2011;20:618–52.
- [17] Wu J-Y. A unified phase-field theory for the mechanics of damage and quasi-brittle failure. *J Mech Phys Solids* 2017;103:72–99.
- [18] Wu J. On the unified phase-field theory for damage and failure in solids and structures: Theoretical and numerical aspects. *Chin J Theoret Appl Mech* 2021;53: 1–29.
- [19] Feng D-C, Wu J-Y. Phase-field regularized cohesive zone model (CZM) and size effect of concrete. *Eng Fract Mech* 2018;197:66–79.
- [20] Wu J-Y. A geometrically regularized gradient-damage model with energetic equivalence. *Comput Methods Appl Mech Eng* 2018;328:612–37.
- [21] Wu J-Y, Nguyen VP. A length scale insensitive phase-field damage model for brittle fracture. *J Mech Phys Solids* 2018;119:20–42.
- [22] Wang Q, Feng Y, Zhou W, Cheng Y, Ma G. A phase-field model for mixed-mode fracture based on a unified tensile fracture criterion. *Comput Methods Appl Mech Eng* 2020;370:113270.
- [23] Wu J-Y, Huang Y, Nguyen VP, Mandal TK. Crack nucleation and propagation in the phase-field cohesive zone model with application to Hertzian indentation fracture. *Int J Solids Struct* 2022;111462.
- [24] Pavoni N, Green R. Recent crustal movements. Elsevier; 2013.
- [25] Sibson RH. Crustal stress, faulting and fluid flow. *Geol Soc Lond Spec Publ* 1994;78:69–84.
- [26] Yilbas B, Arif A, Aleem BA. Laser welding of low carbon steel and thermal stress analysis. *Opt Laser Technol* 2010;42:760–8.
- [27] Satoh K. Thermal stresses developed in high-strength steels subjected to thermal cycles simulating weld heat-affected zone. *Trans Japan Welding Soc* 1972;3: 135–42.
- [28] Bofang Z. Thermal stresses and temperature control of mass concrete. Saint Louis: Elsevier Science & Technology; 2013.
- [29] Xu W, Qiang S, Hu Z, Ding B, Yang B. Effect of hydration heat inhibitor on thermal stress of hydraulic structures with different thicknesses. *Adv Civil Eng* 2020; 2020.
- [30] Ehlers W, Luo C. A phase-field approach embedded in the theory of porous media for the description of dynamic hydraulic fracturing. *Int J Numer Meth Eng* 2017;315:348–68.
- [31] Shiozawa S, Lee S, Wheeler MF. The effect of stress boundary conditions on fluid-driven fracture propagation in porous media using a phase-field modeling approach. *Int J Numer Anal Meth Geomech* 2019;43:1316–40.
- [32] Zhou S, Zhuang X, Rabczuk T. Phase field method for quasi-static hydro-fracture in porous media under stress boundary condition considering the effect of initial stress field. *Theor Appl Fract Mech* 2020;107.
- [33] Salvati E. Residual stress as a fracture toughening mechanism: A Phase-Field study on a brittle material. *Theor Appl Fract Mech* 2021;114.
- [34] Xu W, Li H, Li Y, Wang T, Lu S, Qiang S, et al. A phase field method with adaptive refinement strategy and virtual crack insertion technique. *Eng Fract Mech* 2022;271:108669.
- [35] Xu W, Jiang D, Zhang C, Li H, Qiang S, Li Y, et al. An adaptive mesh refinement strategy for 3D phase modeling of brittle fracture. *Eng Fract Mech* 2023;109241.
- [36] Braides A. Approximation of free-discontinuity problems. Springer Science & Business Media; 1998.
- [37] Ambrosio L, Tortorelli VM. Approximation of functional depending on jumps by elliptic functional via t-convergence. *Commun Pure Appl Math* 1990;43: 999–1036.
- [38] Rao SS. The finite element method in engineering. Butterworth-Heinemann; 2017.
- [39] Wu J-Y, Huang Y, Nguyen VP. On the BFGS monolithic algorithm for the unified phase field damage theory. *Comput Methods Appl Mech Eng* 2020;360:112704.
- [40] Xu W, Li Y, Li H, Qiang S, Zhang C, Zhang C. Multi-level adaptive mesh refinement technique for phase-field method. *Eng Fract Mech* 2022;108891.
- [41] Storvik E, Both JW, Sargado JM, Nordbotten JM, Radu FA. An accelerated staggered scheme for variational phase-field models of brittle fracture. *Comput Methods Appl Mech Eng* 2021;381.
- [42] Goswami S, Anitescu C, Rabczuk T. Adaptive fourth-order phase field analysis for brittle fracture. *Comput Methods Appl Mech Eng* 2020;361:112808.
- [43] Assaf R, Birk C, Natarajan S, Gravenkamp H. Three-dimensional phase-field modeling of brittle fracture using an adaptive octree-based scaled boundary finite element approach. *Comput Methods Appl Mech Eng* 2022;399:115364.
- [44] Xu W, Zhang C, Li Y. Simulation and analysis of rust expansion cracking of reinforced concrete. *Constr Build Mater* 2024;426:136199.